Practice Questions for 2024-11-21-EconNetSimplex

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1 Questions

1. In the economic interpretation of the network simplex method (Slides 1-6), the initial feasible solution is found on a spanning tree. What is the significance of this spanning tree in the context of determining "fair prices" in the network?

   a. The spanning tree ensures that all nodes are connected, allowing for a complete price calculation across the entire network.

   b. The spanning tree defines a set of independent equations, one for each arc, that can be used to solve for the "fair prices" at each node.

   c. The spanning tree guarantees that the resulting "fair prices" will be unique and independent of the starting node.

   d. The spanning tree minimizes the total shipping cost in the network, leading to the most efficient price structure.

   e. The spanning tree ensures that the "fair prices" are consistent with the market prices at the source and sink nodes.

2. In the second step of the network simplex method's economic interpretation (Slide 5), a competitor evaluates potential profit by shipping along a non-tree arc. What condition must be met for this arc to be considered a profitable opportunity for the competitor?

   a. The shipping cost along the non-tree arc must be less than the difference in prices between the two nodes connected by the arc.

   b. The sum of the price at the origin node and the shipping cost along the non-tree arc must be less than the price at the destination node.

   c. The difference between the prices at the two nodes connected by the non-tree arc must be greater than the shipping cost.

   d. The shipping cost along the non-tree arc must be greater than the difference in prices between the two nodes connected by the arc.

   e. The sum of the prices at the two nodes connected by the non-tree arc must be less than the shipping cost.

3. The third step of the network simplex method (Slide 6) involves adjusting the flow in the network after a non-tree arc is selected. What is the key equation that ensures feasibility is maintained after this adjustment?

   a. $y_i + c_{ij} = y_j$

b. $\tilde{y}_j - \tilde{y}_i = c_{ij}$

c. $y_i + c_{ij} < y_j$

d. $\tilde{z}_{ij} < 0$

e. $y_j - y_i + z_{ij} = c_{ij}$

4. In the economic interpretation of the network simplex method, the algorithm iteratively adjusts "fair prices" at each node. What is the ultimate goal of this iterative price adjustment process, and how does it relate to the optimal solution of the network flow problem?

a. To find a set of prices that maximizes the total profit from shipping across the network.

b. To find a set of prices that minimizes the total shipping cost across the network.

c. To find a set of prices that equalizes the prices at all nodes in the network.

d. To find a set of prices that are consistent with the market prices at the source and sink nodes.

e. To find a set of prices such that no further profitable shipping opportunities exist for a competitor.

5. In the economic interpretation of the network simplex method (Slides 1-6), the initial feasible solution is found on a spanning tree. Explain the significance of this spanning tree in the context of determining "fair prices" in the network, and how this relates to the equation $y_i + c_{ij} = y_j$ presented in Slide 4.

a. The spanning tree ensures that all nodes are connected, allowing for a consistent set of prices to be determined using the equation $y_i + c_{ij} = y_j$, which prevents arbitrage opportunities.

b. The spanning tree represents the initial flow pattern, and the equation $y_i + c_{ij} = y_j$ is used to calculate the flow along each arc in the tree.

c. The spanning tree is arbitrary and has no specific significance in determining prices; the equation $y_i + c_{ij} = y_j$ is used to calculate prices regardless of the tree structure.

d. The spanning tree ensures that the network is connected, but the equation $y_i + c_{ij} = y_j$ is only used to calculate prices for arcs not in the spanning tree.

e. The spanning tree is irrelevant to the pricing scheme; the equation $y_i + c_{ij} = y_j$ is used to determine prices based solely on the cost matrix c_{ij}.

6. In the economic interpretation of the network simplex method (Slide 1-6), the initial feasible solution is found on a spanning tree. What is the significance of this spanning tree in the context of determining "fair prices" in the network, and how does this relate to the equation $y_i + c_{ij} = y_j$ presented in Slide 4?

a. The spanning tree ensures that all nodes are connected, allowing for the calculation of prices based on shortest paths from a central node.

b. The spanning tree guarantees a unique path between any two nodes, enabling the consistent application of the price equation $y_i + c_{ij} = y_j$ to determine fair prices based on shipping costs.

c. The spanning tree represents a feasible flow pattern, and the price equation is used to adjust the flow along the arcs of the tree to achieve optimality.

d. The spanning tree is arbitrary and does not have any specific significance in determining fair prices; the price equation is applied independently of the tree structure.

e. The spanning tree ensures that the total supply equals the total demand, which is a necessary condition for the price equation to hold.

7. In the second step of the network simplex method's economic interpretation (Slide 5), a competitor evaluates potential profit by shipping along a non-tree arc. What condition must be met for this arc to be considered a profitable opportunity for the competitor?

   a. The shipping cost along the non-tree arc must be less than the difference in prices between the two nodes connected by the arc.

   b. The sum of the price at the origin node and the shipping cost along the non-tree arc must be less than the price at the destination node.

   c. The difference in prices between the two nodes connected by the non-tree arc must be greater than the shipping cost along the arc.

   d. The price at the destination node must be greater than the price at the origin node, regardless of the shipping cost.

   e. The shipping cost along the non-tree arc must be zero.

8. The third step of the network simplex method (Slide 6) involves adjusting the flow in the network after a non-tree arc is selected. What is the key equation that ensures feasibility is maintained after this adjustment?

   a. $y_i + c_{ij} = y_j$

   b. $\tilde{z}_{ij} = 0$

   c. $\tilde{y}_j - \tilde{y}_i = c_{ij}$

   d. $y_i + c_{ij} < y_j$

   e. There is no specific equation; feasibility is maintained through arbitrary flow adjustments.

9. In the economic interpretation of the network simplex method, the algorithm iteratively adjusts "fair prices" at each node. What is the ultimate goal of this iterative price adjustment process, and how does it relate to the optimal solution of the network flow problem?

   a. The goal is to find prices that minimize the total shipping cost.

   b. The goal is to find prices such that no competitor can find a profitable shipping opportunity along any arc.

   c. The goal is to find prices that maximize the total revenue from shipping.

   d. The goal is to find prices that balance supply and demand at each node.

   e. The goal is to find prices that are equal at all nodes in the network.

10. In the economic interpretation of the network simplex method (Slides 1-6), the initial feasible solution is found on a spanning tree. A competitor then evaluates potential profit by shipping along a non-tree arc (Slide 5). What specific condition, involving the "fair prices" (y_i) and the shipping cost (c_{ij}), must be met for this non-tree arc to be considered a profitable opportunity for the competitor?

    a. $y_i + c_{ij} > y_j$

    b. $y_i + c_{ij} < y_j$

    c. $y_i - c_{ij} > y_j$

    d. $y_i - c_{ij} < y_j$

e. $y_i = y_j + c_{ij}$

2 Answers

1. In the economic interpretation of the network simplex method (Slides 1-6), the initial feasible solution is found on a spanning tree. What is the significance of this spanning tree in the context of determining "fair prices" in the network?

   a. While a spanning tree connects all nodes, this is not the primary reason for its use in determining fair prices. The key is the independence of the equations it provides.

   b. The spanning tree provides a set of $|N| - 1$ independent equations (one for each arc in the tree), where $|N|$ is the number of nodes. These equations, of the form $y_i + c_{ij} = y_j$, allow us to solve for the $|N|$ node prices, up to an additive constant. This is because the price at any node can be expressed as the price at the root node plus the sum of the costs along the path from the root to that node.

   c. The "fair prices" are not unique; they are determined up to an additive constant. The choice of a spanning tree affects the specific values, but not the overall structure of the prices.

   d. The spanning tree used in the initial feasible solution does not necessarily minimize the total shipping cost. The algorithm iteratively improves the solution to minimize cost.

   e. The spanning tree does not directly ensure consistency with market prices at the source and sink. The algorithm adjusts prices iteratively to reach an optimal solution.

2. In the second step of the network simplex method's economic interpretation (Slide 5), a competitor evaluates potential profit by shipping along a non-tree arc. What condition must be met for this arc to be considered a profitable opportunity for the competitor?

   a. This condition is incorrect. It doesn't consider the price at the origin node.

   b. The condition $y_i + c_{ij} < y_j$ indicates a profitable opportunity. If the cost of buying at node i and shipping to node j is less than the selling price at node j, the competitor can make a profit.

   c. This condition is equivalent to the correct answer but is not the way it is presented in the slides.

   d. This condition would indicate an unprofitable opportunity.

   e. This condition is irrelevant to the profitability of shipping along the arc.

3. The third step of the network simplex method (Slide 6) involves adjusting the flow in the network after a non-tree arc is selected. What is the key equation that ensures feasibility is maintained after this adjustment?

   a. This equation is used in the first step to determine initial prices, not to maintain feasibility after adjustments.

   b. The equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ ensures that the adjusted prices ($\tilde{y}_i$ and $\tilde{y}_j$) are consistent with the shipping cost (c_{ij}) along the arcs in the updated spanning tree. This maintains feasibility.

   c. This inequality is used in the second step to identify profitable opportunities, not to maintain feasibility.

   d. This inequality indicates a negative reduced cost, which is a condition for selecting a non-tree arc, not for maintaining feasibility.

   e. This equation is used in the second step to relate prices and costs, not to maintain feasibility after adjustments.

4. In the economic interpretation of the network simplex method, the algorithm iteratively adjusts "fair prices" at each node. What is the ultimate goal of this iterative price adjustment process, and how does it relate to the optimal solution of the network flow problem?

   a. The algorithm minimizes cost, not maximizes profit. Profit is implicitly considered through the cost minimization.

   b. While the algorithm minimizes cost, this is not the direct goal of the price adjustment. The price adjustment ensures optimality, which leads to minimum cost.

   c. The prices are not equalized at all nodes. The price differences reflect the shipping costs and the optimal flow pattern.

   d. The algorithm does not explicitly aim for consistency with market prices at the source and sink. The optimal solution implicitly incorporates these factors.

   e. The iterative price adjustment aims to eliminate any profitable shipping opportunities for a competitor. When no such opportunities exist ($y_i + c_{ij} \geq y_j$ for all arcs), the current solution is optimal, and the prices reflect the optimal flow pattern in the network.

5. In the economic interpretation of the network simplex method (Slides 1-6), the initial feasible solution is found on a spanning tree. Explain the significance of this spanning tree in the context of determining "fair prices" in the network, and how this relates to the equation $y_i + c_{ij} = y_j$ presented in Slide 4.

   a. The spanning tree ensures connectivity, allowing for a consistent set of prices to be determined. The equation $y_i + c_{ij} = y_j$ is applied to each arc in the tree, establishing a relationship between prices at connected nodes. This prevents arbitrage, where a commodity could be bought at a low price and sold at a higher price at another node, resulting in a profit greater than the shipping cost. The prices are "fair" in the sense that they reflect the cost of transportation between nodes.

   b. While the spanning tree represents the initial flow, the equation $y_i + c_{ij} = y_j$ is used to determine prices, not flow. Flow is determined later in the algorithm.

   c. The spanning tree is not arbitrary; it is crucial for establishing a consistent pricing scheme. The equation $y_i + c_{ij} = y_j$ is applied to arcs within the tree to determine prices.

   d. The equation $y_i + c_{ij} = y_j$ is applied to arcs within the spanning tree to establish initial prices. Arcs not in the tree are considered later in the algorithm.

   e. The spanning tree is essential for establishing a consistent pricing scheme. The equation $y_i + c_{ij} = y_j$ is applied to the arcs in the tree, and the cost matrix c_{ij} is a crucial input to the pricing scheme.

6. In the economic interpretation of the network simplex method (Slide 1-6), the initial feasible solution is found on a spanning tree. What is the significance of this spanning tree in the context of determining "fair prices" in the network, and how does this relate to the equation $y_i + c_{ij} = y_j$ presented in Slide 4?

   a. While the spanning tree connects all nodes, the significance is not about shortest paths but about the unique path between any two connected nodes. Shortest paths are not relevant to the initial price determination.

   b. The spanning tree's significance lies in its acyclicity, guaranteeing a unique path between any pair of nodes connected in the tree. This uniqueness is crucial for consistently applying the price equation $y_i + c_{ij} = y_j$ along each arc (i, j) in the tree. Inconsistent application would lead to price discrepancies and arbitrage opportunities. The equation ensures that the price difference between two nodes reflects the exact shipping cost along the unique path connecting them within the tree.

   c. The spanning tree represents an initial feasible solution, but the price equation is used to determine prices, not to directly adjust the flow. Flow adjustments happen later in the algorithm.

d. The spanning tree is not arbitrary; its structure is fundamental to the initial price determination because it defines the unique paths used in the price equation.

e. While supply and demand balance is important for the overall problem, the spanning tree's role in price determination is independent of this balance. The price equation is applied locally along the arcs of the tree.

7. In the second step of the network simplex method's economic interpretation (Slide 5), a competitor evaluates potential profit by shipping along a non-tree arc. What condition must be met for this arc to be considered a profitable opportunity for the competitor?

a. This is incorrect. The condition is that the price difference must be greater than the shipping cost, not less than.

b. This is the inverse of the correct condition. The sum of the origin price and shipping cost must be *less* than the destination price for profit.

c. A profitable opportunity for the competitor exists if the price difference between the destination and origin nodes exceeds the shipping cost. This means the competitor can buy at a lower price, ship, and sell at a higher price, making a profit.

d. This is insufficient. The price difference must exceed the shipping cost to ensure profit.

e. This is not a necessary condition. Profit can be made even with positive shipping costs if the price difference is large enough.

8. The third step of the network simplex method (Slide 6) involves adjusting the flow in the network after a non-tree arc is selected. What is the key equation that ensures feasibility is maintained after this adjustment?

a. This equation is used in the first step to determine initial fair prices, not to maintain feasibility during flow adjustments.

b. While $\tilde{z}_{ij} = 0$ is a consequence of maintaining feasibility, it's not the key equation used to *ensure* feasibility during the adjustment process.

c. The equation $\tilde{y}_j - \tilde{y}_i = c_{ij}$ ensures that after the flow adjustment, the price difference between nodes i and j exactly reflects the shipping cost c_{ij}. This condition is equivalent to $\tilde{z}_{ij} = 0$, which means there is no incentive for further flow adjustments along the arc (i, j).

d. This inequality is used in the second step to identify a profitable non-tree arc, not to maintain feasibility during flow adjustments.

e. The flow adjustments are not arbitrary; they are carefully calculated to maintain feasibility and optimality.

9. In the economic interpretation of the network simplex method, the algorithm iteratively adjusts "fair prices" at each node. What is the ultimate goal of this iterative price adjustment process, and how does it relate to the optimal solution of the network flow problem?

a. Minimizing total shipping cost is the goal of the network flow problem itself, but the price adjustment process aims to identify the conditions under which this minimum is achieved.

b. The iterative price adjustment aims to reach a state where no further profitable shipping opportunities exist for any competitor. This condition implies that the current flow pattern is optimal, as any deviation would be unprofitable. The absence of arbitrage opportunities is directly linked to the optimality of the network flow solution.

c. Maximizing revenue is not the direct goal of the price adjustment process; it's a consequence of finding the optimal flow pattern.

d. Balancing supply and demand is a constraint of the problem, but the price adjustment process focuses on eliminating arbitrage opportunities, not directly on supply-demand balance.

e. Equal prices at all nodes would imply zero shipping costs and is not the general goal; prices will differ to reflect shipping costs.

10. In the economic interpretation of the network simplex method (Slides 1-6), the initial feasible solution is found on a spanning tree. A competitor then evaluates potential profit by shipping along a non-tree arc (Slide 5). What specific condition, involving the "fair prices" (y_i) and the shipping cost (c_{ij}), must be met for this non-tree arc to be considered a profitable opportunity for the competitor?

    a. This inequality suggests that the cost of shipping exceeds the profit, making the arc unprofitable for a competitor.

    b. The condition $y_i + c_{ij} < y_j$ indicates a profitable opportunity. If the cost of buying at node i (y_i) and shipping to node j (c_{ij}) is less than the selling price at node j (y_j), a competitor can profit by shipping along this arc.

    c. This inequality is not relevant to the economic interpretation of the network simplex method in this context.

    d. This inequality is not relevant to the economic interpretation of the network simplex method in this context.

    e. This equation represents the condition for "fair prices" along arcs in the spanning tree, not a profitable opportunity for a competitor along a non-tree arc.